Electric Vehicles & Traction

A professional engineering handbook on EV relevance, vehicle motion, conventional and electric powertrains, traction systems, speed-time curves, EV motors, energy storage hybridization, HEV control architecture, EV charging standards, public charging infrastructure and e-mobility.

What this subject is

This course combines electric vehicle fundamentals with electric traction. Students learn vehicle motion resistance, tractive effort, IC and EV powertrains, traction supply and motors, BLDC/PMSM drives, batteries, ultra-capacitors, flywheels, fuel cells, hybridized storage and EV charging infrastructure.

Malayalam: EV traction design-□ battery □□□□□□□ □□□□□□□□□□□ □□□□□. Vehicle resistance, rolling resistance, aerodynamic drag, motor torque, battery power, controller, charger, connector safety □□□□□□□ □□□□□ system □□□ □□□□□□□□□□□□□□□□.

Official syllabus information

Item	Value
Program	Diploma in Electrical and Electronics Engineering
Course code	6032D
Course title	Electric Vehicles & Traction
Semester	6
Credits	4
Category	Program Elective
Periods	4 per week (L:3 T:1 P:0); 60 per semester
CO-PO Mapping	CO1 moderately mapped to PO1 and strongly to PO5; CO2, CO3 and CO4 moderately mapped to PO1

Course objectives

- Express relevance of EVs for future transportation.

- Illustrate electric traction fundamentals and EV motors.
- Outline EV energy storage options.
- Identify charging infrastructure requirements.

Prerequisite

Basics of Electrical Circuits from Fundamentals of Electrical & Electronics Engineering, Semester 2.

Course outcomes

- CO1: Outline EV basic concepts and powertrain needs.
- CO2: Explain electric traction and EV motor applications.
- CO3: Outline hybridization of EV energy storage.
- CO4: Summarize public charging infrastructure and e-mobility.

Redesigned syllabus roadmap

The official syllabus converted into a practical EV traction engineering route.

Module 1 · CO1 · 15 Hours

EV Basics, Vehicle Motion and Powertrain Needs

Automobile development, EV/HEV/PHEV importance, resistance forces, tractive effort and comparison of conventional and electric powertrains.

Automobile development and EV relevance

Conventional gasoline vehicles use fuel combustion to produce shaft power through engine, clutch, gearbox and differential. EVs use electrical energy stored in a battery or generated through a fuel cell and convert it to wheel torque through motor drives. HEV and PHEV combine engine and electric propulsion to improve efficiency and reduce emissions.

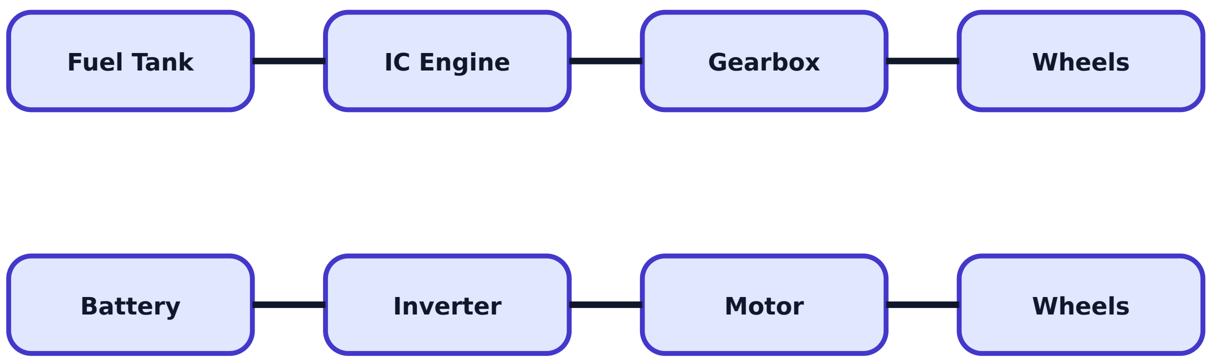
Malayalam: Gasoline vehicle-□ energy path fuel → engine → gearbox → wheels □□□. EV-□ battery → inverter/controller → motor → wheels. HEV/PHEV-□ engine and motor □□□□□□ power source □□□ □□□□.

Vehicle movement factors

Vehicle motion is opposed by rolling resistance, aerodynamic drag, gradient resistance and acceleration resistance. Tyre-ground adhesion limits how much tractive effort can be transferred without wheel slip. EV powertrain selection must match torque, speed, gradeability and acceleration needs.

Exam tip: For tractive effort questions, list every resistance separately before writing final total force.

Powertrain comparison



IC and electric powertrain

An IC powertrain needs clutch, gearbox and differential because the engine operates efficiently over limited speed range. EV motor drives can provide high starting torque and wide speed control, so many EVs use simpler single-speed reduction. HEV and PHEV powertrains add energy management to decide power split between engine, motor and battery.

EV, HEV and PHEV comparison

Type	Energy source	Charging	Key feature
EV	Battery electric	External charging	Zero tail-pipe emission
HEV	Fuel + battery	Mostly internal regeneration/engine	Better fuel economy
PHEV	Fuel + larger battery	External charging possible	Electric-only short range plus engine backup

Expected questions

1. Explain social and environmental importance of EV and HEV.
2. Explain vehicle resistance, rolling resistance, aerodynamic drag and grading resistance.
3. Explain tractive effort on wheels.
4. Draw and explain IC engine powertrain.
5. Compare EV, HEV and PHEV powertrains.

Module 2 · CO2 · 15 Hours

Electric Traction and EV Motor Applications

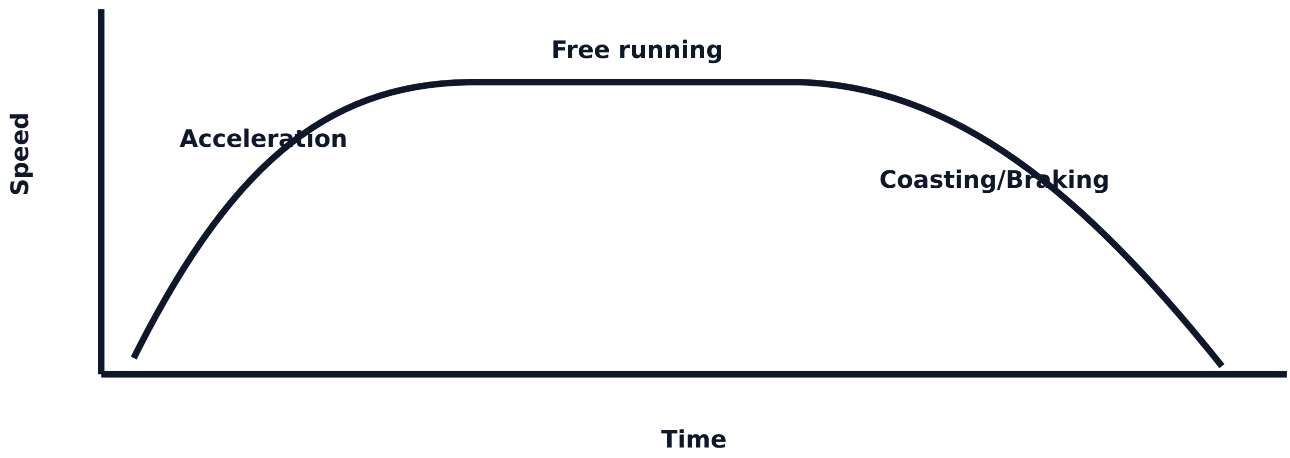
Traction systems, supply layout, ideal traction motor requirements, speed-time curve, hybrid drivetrains, BLDC and PMSM motors.

Electric traction basics

Electric traction uses electric motors for propulsion of trains, trams, metros and electric vehicles. It offers high starting torque, smooth acceleration, regenerative braking, low local emission and good controllability. An ideal traction system should provide high torque at start, wide speed range, robust operation, high efficiency and reliable braking.

Malayalam: Traction motor-□□ starting torque □□□□□□□□□ speed control smooth □□□□ □□□□. Train/EV load sudden □□□ □□□□□□□□□□□ motor reliable □□□□□□□□□□.

Typical speed-time curve



Traction supply and motors

The syllabus includes traction supply systems, specifications and layout of electric locomotive system. Traction motors listed are DC series motor, single phase AC series motor and three phase induction motor. Requirements include high starting torque, overload capacity, speed control, ruggedness and braking ability.

Hybrid drive train configurations

Hybrid electric powertrain design may be series, parallel or series-parallel. Series uses engine-generator set to supply electrical power; only motor drives wheels. Parallel allows engine and motor both to supply mechanical torque. Series-parallel combines both power paths for better flexibility.

BLDC and PMSM motors

Brushless DC motors use electronic commutation instead of brushes, improving reliability and efficiency. PMSM uses permanent magnet rotor and AC stator currents synchronized with rotor position. Both are important EV motors due to power density, efficiency and controllability.

> **Industrial note:** In EV service, motor is not selected by name only. Check torque-speed curve, cooling, controller compatibility, duty cycle and regenerative braking support.

Expected questions

1. Explain electric traction and advantages.
2. List requirements of ideal traction system and traction motors.
3. Draw and explain speed-time curve.
4. Explain series, parallel and series-parallel hybrid drive trains.
5. Explain BLDC and PMSM motor working with diagram.

Module 3 · CO3 · 14 Hours

Energy Storage and Hybridization in EV

Battery parameters, lithium batteries, ultra-capacitors, flywheels, hydrogen fuel cells and hybridization of storage devices.

Battery parameters

Battery performance is described by voltage, capacity, energy density, specific power, efficiency, thermodynamic voltage, SOC, cycle life, safety and temperature performance. Selection of EV battery depends on range, acceleration, charging time, life, cost, safety and available packaging space.

Malayalam: Battery Ah □□□□□□ □□□□□□□□□. Energy, specific power, charging rate, life, heating, safety, cost □□□□□□ EV design-□ □□□□□□ matter □□□□□□□□□□.

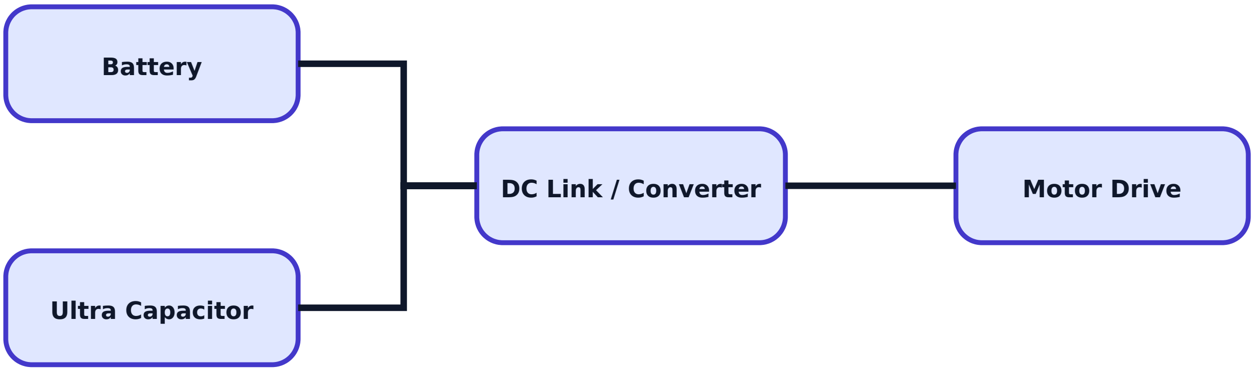
Lithium batteries

Lithium-ion and lithium polymer batteries are common EV storage options because of high energy density and good cycle performance. Limitations include cost, thermal safety concerns, need for BMS, ageing and recycling challenges.

Storage	Strength	Limitation
Li-ion battery	High energy density	Needs BMS and thermal care

Ultra-capacitor	Very high power, fast charge	Low energy density
Flywheel	Fast energy exchange	Mechanical complexity
Fuel cell	Long range potential	Hydrogen infrastructure and cost

Hybridized storage concept



Ultra-capacitors and flywheels

Ultra-capacitors are suitable for rapid charge/discharge during acceleration support and regenerative braking. Flywheels store energy mechanically as rotational kinetic energy and can deliver high power quickly, but require safety containment and precise bearings.

Fuel cells and hybridization

Fuel cells convert chemical energy of fuel directly into electricity. Hydrogen fuel cells produce electricity with water as the main product, but need hydrogen storage and refuelling infrastructure. Hybridization combines battery, ultra-capacitor, flywheel or fuel cell to balance energy and power demands.

Expected questions

1. Define specific power, energy efficiency and thermodynamic voltage.
2. Explain lithium-ion and lithium polymer batteries.
3. Explain ultra-capacitor and flywheel role in EV.
4. Explain hydrogen fuel cell working, advantages and limitations.
5. Explain need and importance of energy storage hybridization.

Module 4 · CO4 · 14 Hours

HEV Control, Charging Standards and E-Mobility

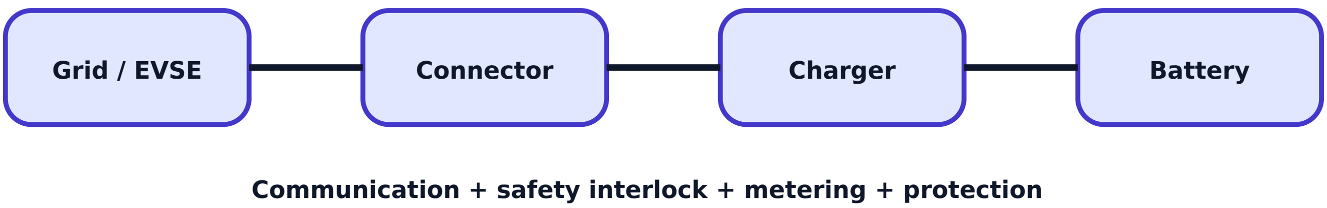
HEV control architecture, EV charging system, SAE J1772 modes, connectors, V2G/G2V/V2B/V2H, public charging requirements and e-mobility challenges.

HEV control architecture

HEV control coordinates engine, motor, battery, inverter, DC-DC converter, charger and vehicle controller. It decides power split, regenerative braking, engine start/stop, battery protection and driver torque request.

Malayalam: HEV controller vehicle brain □□□□ □□□. Driver accelerator/brake input, battery SOC, motor limit, engine condition □□□□□□□ □□□□□□□ power □□□□□ □□□□□□□ □□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□□□□□.

EV charging system block



Charging standards and connectors

EV charging includes different charging types and modes as per SAE J1772. The syllabus includes V2G, G2V, V2B, V2H, connector requirements, input-output and safety. Connector systems include CCS, Bharat EV AC and Bharat EV DC. Important requirements are current rating, voltage rating, earthing, insulation, communication, locking and environmental protection.

Public charging infrastructure

Public charging infrastructure requires reliable electrical supply, transformer capacity, protection, metering, charger management system, user authentication, parking layout, signage, safety clearance, emergency shutdown and maintenance support. Operational requirements include billing, uptime, queue control and remote monitoring.

E-mobility scope and challenges

E-mobility reduces oil dependency and local emissions, supports clean transport and opens new service industries. Challenges include battery cost, charging availability, grid loading, standardization, battery recycling, safety, skilled maintenance and user adoption.

Expected questions

1. Draw HEV control architecture block diagram.
2. Explain EV charging system with block diagram.
3. Explain SAE J1772 charging modes and connector requirements.
4. Explain CCS, Bharat EV AC and Bharat EV DC.
5. Summarize public charging infrastructure requirements and e-mobility challenges.

Formula / Rule Bank

EV traction formulas, storage rules and charging calculations.

Rolling resistance

Fr = coefficient of rolling resistance x vehicle weight.

Aerodynamic drag

Fd = 0.5 x air density x Cd x frontal area x velocity squared.

Grade resistance

Fg = vehicle weight x sin(theta), approximately weight x gradient for small angle.

Tractive effort

Total tractive effort = rolling + aerodynamic + grade + acceleration resistance.

Average speed

Average speed = distance / actual running time.

Scheduled speed

Scheduled speed = distance / total time including stops.

Battery energy

Energy (Wh) = voltage x ampere-hour capacity.

Charging time

Charging time = energy to be charged / charger power, corrected for efficiency.

Solved Examples / Problem Lab

Step-by-step EV traction and charging examples.

Example 1 - Battery energy

Given: Battery pack voltage = 320 V, capacity = 100 Ah.

Energy = V x Ah = 320 x 100 = 32000 Wh.

Battery energy = 32 kWh.

Example 2 - Charging time

Energy to charge = 24 kWh, charger power = 6 kW.

Time = 24 / 6.

Ideal charging time = 4 hours.

Example 3 - Average speed

Distance = 40 km, running time = 1 hour.

Average speed = 40 km/h.

Example 4 - Scheduled speed

Distance = 40 km, total time including stops = 1.25 h.

Scheduled speed = 40/1.25 = 32 km/h.

Example 5 - Tractive effort concept

If rolling resistance = 250 N, aerodynamic drag = 180 N and grade resistance = 400 N.

Total tractive effort ignoring acceleration = 830 N.

Practice Section and Expected Questions

Module-wise exam preparation for EV and traction.

One-mark questions

1. Define tractive effort.
2. What is PHEV?

3. What is crest speed?
4. Name two special motors used in EV.
5. What is V2G?

Short-answer questions

1. List factors affecting vehicle motion.
2. List requirements of ideal traction system.
3. Compare average speed and scheduled speed.
4. List battery parameters.
5. List public charging infrastructure requirements.

Descriptive questions

1. Compare EV, HEV and PHEV powertrains.
2. Explain speed-time curve and its importance.
3. Explain BLDC and PMSM working.
4. Explain hybridization of energy storage devices.
5. Explain EV charging standards, connectors and e-mobility challenges.

MCQ practice

1. Scheduled speed includes: A) stops B) only acceleration C) only braking D) battery SOC only

2. PMSM rotor uses: A) carbon brush only B) permanent magnets C) water pump D) fuse wire

3. G2V means: A) Grid to Vehicle B) Gear to Voltage C) Ground to Vehicle D) Generator to Valve

Fresh Model Question Paper

Original model paper based on syllabus coverage; not copied from official question papers.

Course 6032D - Electric Vehicles & Traction

Time: 3 Hours **Maximum Marks:** 100

Part A - Short Answer

1. Define electric vehicle.
2. What is rolling resistance?
3. What is scheduled speed?
4. Name any two EV energy storage devices.
5. What is CCS connector?

Part B - Medium Answer

1. Compare gasoline vehicle, EV, HEV and PHEV.
2. Explain vehicle resistance and tractive effort.
3. Explain electric traction and requirements of traction motors.
4. Explain battery parameters and lithium batteries.
5. Explain EV charging modes and connector safety requirements.

Part C - Long Answer / Problems

1. Draw and explain conventional and electric vehicle powertrains.
2. Draw and explain speed-time curve for passenger service.
3. Explain BLDC and PMSM motor working with diagrams.
4. A 320 V, 100 Ah battery is charged using a 6 kW charger. Calculate battery energy and ideal charging time.
5. Explain HEV control architecture, public charging infrastructure and challenges of e-mobility.

Complete Model Answers

Answer guide with formulas, diagram points and common mistakes.

EV definition answer

An EV is a vehicle propelled by one or more electric motors using electrical energy stored in a battery or supplied by another electric source.

Battery numerical answer

Energy = V x Ah = 320 x 100 = 32000 Wh = 32 kWh. Charging time = 32 / 6 = 5.33 h ideal.

Battery energy = 32 kWh; ideal charging time ≈ 5.33 hours.

Speed-time curve answer points

Draw speed on vertical axis and time on horizontal axis. Mark acceleration, free running, coasting and braking. Mention average speed, scheduled speed and crest speed.

Motor answer points

BLDC uses electronic commutation and rotor position sensing; PMSM uses permanent magnet rotor with synchronized AC stator field. Mention efficiency, torque control and regenerative braking suitability.

Common mistakes

Do not write EV as only battery plus motor. Include controller, inverter, charger, BMS, converter, motor, drivetrain, protection and communication.

References from syllabus

Ehsani, B.L. Theraja, Iqbal Husain, A.K. Babu, Fuhs, Gianfranco, Chan and Chau, Lechner, Rashid, Moorthi, NPTEL, FAME, ANERT, Ministry of Power and KSEB resources.